

Radiative transitions of strange heavy axial mesons in light-cone QCD sum rules

S. Bilmis,¹ T. M. Aliev,¹ and M. Savci¹

¹*Department of Physics, Middle East Technical University, Ankara, Turkey*

(Dated: July 3, 2026)

Abstract

We study the radiative transitions $D_{s1}(2460) \rightarrow D_s\gamma$, $D_{s1}(2536) \rightarrow D_s\gamma$, and $B_{s1}(5830) \rightarrow B_s\gamma$ in light-cone QCD sum rules. The calculation keeps strange-quark-mass terms, hard photon emission, and two- and three-particle photon distribution-amplitude contributions. With the lattice-normalized photon distribution amplitude we obtain $\Gamma[D_{s1}(2460) \rightarrow D_s\gamma] = 16.1^{+4.4}_{-3.7}$ keV and $\Gamma[D_{s1}(2536) \rightarrow D_s\gamma] = 0.50^{+0.60}_{-0.38}$ keV. The strong suppression of the latter mode follows from destructive interference between the two physical-current components.

I. INTRODUCTION

The spectroscopy of positive-parity strange heavy mesons remains a useful testing ground for the interplay between heavy-quark symmetry, light-quark dynamics, and nearby hadronic thresholds. The charm-strange states $D_{s1}(2460)$ and $D_{s1}(2536)$ are especially interesting. The lower state is very narrow and has observed radiative decay modes, whereas the upper state is seen clearly in hadronic channels but its radiative transition to $D_s\gamma$ has not been observed with a comparable significance [1–3]. In the bottom-strange sector the state $B_{s1}(5830)$ has been established in hadronic spectroscopy [4], while direct radiative information is still absent.

Radiative widths of these mesons have been considered in quark models, relativistic quark models, effective approaches, and phenomenological analyses [5–15]. A recent observation by Bondar and Milstein is particularly relevant for the present work: the smallness of $D_{s1}(2536) \rightarrow D_s\gamma$ can be understood as a cancellation between the two spin components of the physical axial state [14, 15]. One of our aims is to examine the same mechanism in a light-cone QCD sum-rule (LCSR) calculation.

The LCSR method expresses the correlation function near the light cone in terms of photon distribution amplitudes (DAs) and hadronic decay constants [16–19]. It is well suited for exclusive radiative transitions because it keeps the hard photon emission from quark lines and the long-distance photon emission encoded in photon DAs in a single framework. In the present analysis we include both quark-line hard emission terms, the two-particle photon DAs, and the three-particle photon-DA terms induced by the background-gluon insertion in the quark propagator. We also keep the terms proportional to the strange-quark mass.

The paper is organized as follows. In Sec. II we define the currents, the radiative form

factor, and the LCSR correlation function. The sum-rule derivation and the physical-current normalization are summarized in Sec. III. Section IV contains the input parameters, Borel-window discussion, numerical results, comparison with other studies, and experimental interpretation. We conclude in Sec. V.

II. THEORETICAL FRAMEWORK

We consider the transition of an axial strange heavy meson with momentum $P = p + q$ into a pseudoscalar strange heavy meson with momentum p and a real photon with momentum q . In mass and charge labels, the subscript q denotes the light quark and $Q = c, b$ the heavy quark; for the present applications $q = s$. The standalone four-vector q^μ continues to denote the photon momentum. The pseudoscalar interpolating current is

$$J_5 = \bar{q} i \gamma_5 Q. \quad (1)$$

For the axial channel we use the two basis currents

$$J_\mu^A = \bar{q} \gamma_\mu \gamma_5 Q, \quad (2)$$

$$J_\mu^B = \frac{i}{m_Q + m_q} \bar{q} \sigma_{\mu\alpha} P^\alpha \gamma_5 Q, \quad (3)$$

where P^α is the momentum of the initial axial meson. The physical currents are written as

$$J_{1\mu} = \sin \theta J_\mu^A + \cos \theta J_\mu^B, \quad (4)$$

$$J_{2\mu} = \cos \theta J_\mu^A - \sin \theta J_\mu^B. \quad (5)$$

Unless stated otherwise we use $\theta = 35.3^\circ$ and include the interval $25^\circ \leq \theta \leq 45^\circ$ as a systematic scan.

Following the notation commonly used in the LCSR analysis of radiative D_{sJ} transitions [19], the hadronic parameters are defined by

$$\langle 0 | \bar{q} i \gamma_5 Q | P(p) \rangle = \frac{m_P^2 f_P}{m_Q + m_q}, \quad (6)$$

$$\langle 0 | J_{1\mu} | D_{s1}(2460)(P, \eta) \rangle = f_1 m_1 \eta_\mu, \quad (7)$$

$$\langle 0 | J_{2\mu} | D_{s1}(2536)(P, \eta) \rangle = f_2 m_2 \eta_\mu. \quad (8)$$

Here $m_1 = m_{D_{s1}(2460)}$ and $m_2 = m_{D_{s1}(2536)}$, and f_1 and f_2 are the leptonic decay constants of the corresponding $D_{s1}(2460)$ and $D_{s1}(2536)$ mesons. The LCSR OPE is first derived for the basis currents J_μ^A and J_μ^B , and only afterwards rotated to the physical 1^+ states.

For a physical axial meson $X = D_{s1}(2460), D_{s1}(2536), B_{s1}(5748)$, or $B_{s1}(5830)$, the radiative matrix element is parametrized by a single gauge-invariant electric-dipole form factor:

$$\langle P(p)\gamma(q, \varepsilon) | X(P, \eta) \rangle = eg_X [(\eta \cdot \varepsilon^*)(p \cdot q) - (\eta \cdot q)(p \cdot \varepsilon^*)]. \quad (9)$$

The corresponding width is

$$\Gamma(X \rightarrow P\gamma) = \frac{\alpha_{\text{em}}}{3} g_X^2 \left(\frac{m_X^2 - m_P^2}{2m_X} \right)^3. \quad (10)$$

III. LIGHT-CONE SUM RULE

The starting point is the vacuum-to-photon correlation function for the physical mixed current,

$$\Pi_{i\mu}(p, q) = i \int d^4x e^{ip \cdot x} \langle \gamma(q, \varepsilon) | T \{ J_{i\mu}(x) \bar{Q}(0) i\gamma_5 q(0) \} | 0 \rangle, \quad i = 1, 2, \quad (11)$$

where $J_{1\mu}$ and $J_{2\mu}$ are defined in Eq. (5). Since these currents are linear combinations of J_μ^A and J_μ^B , the OPE is evaluated for the two basis correlators and then combined with the same mixing coefficients. We project the correlator onto

$$(p \cdot q)\varepsilon_\mu^* - (p \cdot \varepsilon^*)q_\mu, \quad (12)$$

which matches the structure in Eq. (9).

For later use we write the invariant decomposition as

$$\Pi_{i\mu}(p, q) = \Pi_i(p^2, P^2) [(p \cdot q)\varepsilon_\mu^* - (p \cdot \varepsilon^*)q_\mu] + \Pi_i^{\text{other}} q_\mu + \dots, \quad i = 1, 2. \quad (13)$$

Only the coefficient Π_i of the gauge-invariant E1 structure is used in the sum rule. The basis amplitudes Π_A and Π_B are defined analogously by replacing $J_{i\mu}$ with J_μ^A and J_μ^B , respectively.

On the hadronic side, isolating the ground-state double pole gives

$$\Pi_{i\mu}^{\text{had}} = \frac{m_P^2 f_P}{m_Q + m_q} \frac{m_i f_i g_i}{(m_i^2 - P^2)(m_P^2 - p^2)} [(p \cdot q)\varepsilon_\mu^* - (p \cdot \varepsilon^*)q_\mu] + \dots, \quad i = 1, 2, \quad (14)$$

where (m_i, f_i, g_i) denote the mass, leptonic decay constant, and radiative form factor of the meson coupled to $J_{i\mu}$. The ellipsis denotes higher states and continuum contributions. With our convention $[f_P] = [f_i] = \text{GeV}$ and $[g_i] = \text{GeV}^{-1}$; therefore the prefactor in Eq. (14) has

dimension GeV^{-1} , and after multiplication by the tensor structure of dimension GeV^2 the correlator has dimension GeV , as required.

The double Borel transformation and continuum subtraction are performed in the variables P^2 and p^2 . In the symmetric point used numerically,

$$M_1^2 = M_2^2 = 2M^2, \quad u_0 = \frac{M_1^2}{M_1^2 + M_2^2} = \frac{1}{2}, \quad (15)$$

so the common exponential weight in the spectral integrals is $\exp(-s/M^2)$.

The QCD side is obtained by expanding near the light cone in the external electromagnetic field. We use the quark propagators in the external photon and gluon background. For the heavy-quark line the relevant terms are

$$\begin{aligned} S_Q(k) = & \frac{\not{k} + m_Q}{m_Q^2 - k^2} - e_Q \int_0^1 du \left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2} F^{\mu\nu}(ux) \sigma_{\mu\nu} + \frac{ux_\mu F^{\mu\nu}(ux) \gamma_\nu}{m_Q^2 - k^2} \right] \\ & - g_s \int_0^1 du \left[\frac{\not{k} + m_Q}{2(m_Q^2 - k^2)^2} G^{\mu\nu}(ux) \sigma_{\mu\nu} + \frac{ux_\mu G^{\mu\nu}(ux) \gamma_\nu}{m_Q^2 - k^2} \right] + \dots \end{aligned} \quad (16)$$

The light-quark propagator is used consistently in the same external-field expansion. Ordinary vacuum-condensate terms are not inserted as independent three-point contributions in the photon LCSR. In this calculation their role is replaced by vacuum-to-photon matrix elements such as $\langle \gamma | \bar{q} \sigma_{\mu\nu} q | 0 \rangle$, whose normalization contains the susceptibility combination $\chi_q \langle \bar{q} q \rangle$ or, equivalently, the modern lattice normalization $f_{\gamma,q}^\perp$.

For completeness we spell out the invariant amplitudes used in the numerical analysis. The explicit expressions below are written at the symmetric Borel point used in the calculation, $u_0 = M_1^2/(M_1^2 + M_2^2) = 1/2$, with the effective Borel parameter denoted by M^2 . We use Rohrwild's notation for the photon DAs: the two-particle functions are denoted by $\phi_\gamma(u)$, $\psi^{(v)}(u)$, $\mathcal{A}(u)$ and $\mathcal{B}(u)$, while the three-particle twist-4 DAs are $\mathcal{S}$, $\tilde{\mathcal{S}}$ and $\mathcal{T}_{1,\dots,4}$ [20]. Their explicit forms and numerical parameters are collected in Appendix A. In this notation

the axial-current invariant amplitude is decomposed as

$$\begin{aligned}
\widehat{\Pi}_A = & \int_{(m_Q+m_q)^2}^{s_0} ds e^{-s/M^2} \rho_A(s) + e_Q e^{-m_Q^2/M^2} \langle \bar{q}q \rangle \left[1 - \frac{m_Q m_q}{M^2} + \frac{m_q^2}{2M^2} \left(1 + \frac{m_Q^2}{M^2} \right) \right] \\
& + e_q \langle \bar{q}q \rangle \left(e^{-m_Q^2/M^2} - e^{-s_0/M^2} \right) M^2 \chi_q \phi_\gamma(u_0) \\
& - e_q \langle \bar{q}q \rangle e^{-m_Q^2/M^2} \left[-\frac{1}{4} (\mathcal{A}(u_0) + \mathcal{B}(u_0)) \left(1 + \frac{m_Q^2}{M^2} \right) \right] \\
& + 2e_q f_{3\gamma} m_Q e^{-m_Q^2/M^2} \Psi^{(v)}(u_0) \\
& + e_q \langle \bar{q}q \rangle e^{-m_Q^2/M^2} \mathcal{I}_{F_1}(u_0).
\end{aligned} \tag{17}$$

Here

$$\mathcal{B}(u_0) = -8 \int_0^{u_0} du (u_0 - u) h_\gamma(u), \quad \Psi^{(v)}(u_0) = \int_0^{u_0} du \psi^{(v)}(u). \tag{18}$$

The first relation is only a notation dictionary: it rewrites the two-particle twist-4 combination used in the numerical code, $\mathcal{A} - 8\bar{H}_\gamma$, as Rohrwild's $\mathcal{A} + \mathcal{B}$. The three-particle photon-DA integral is

$$\begin{aligned}
\mathcal{I}_{F_1}(u_0) = & \int_0^{1-u_0} dv \int_0^{u_0/(1-v)} d\alpha_g F_1(u_0 - (1-v)\alpha_g, 1 - u_0 - v\alpha_g, \alpha_g, v) \\
& + \int_{1-u_0}^1 dv \int_0^{(1-u_0)/v} d\alpha_g F_1(u_0 - (1-v)\alpha_g, 1 - u_0 - v\alpha_g, \alpha_g, v),
\end{aligned} \tag{19}$$

with

$$F_1 = \mathcal{S} + \tilde{\mathcal{S}} - \mathcal{T}_1 - \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4 + 2v(-\mathcal{S} - \mathcal{T}_3 + \mathcal{T}_2). \tag{20}$$

The perturbative spectral density in Eq. (17) is

$$\rho_A(s) = e_q \mathcal{R}(s, m_q, m_Q, m_q, m_q^2(m_Q - m_q)) - e_Q \mathcal{R}(s, m_Q, m_q, m_Q, m_Q^2(m_q - m_Q)), \tag{21}$$

where

$$\begin{aligned}
\mathcal{R}(s, m_i, m_j, a, b) = & -\frac{3}{8\pi^2} \left[2a \ln \frac{s - m_j^2 + m_i^2 - \lambda^{1/2}(s, m_j^2, m_i^2)}{s - m_j^2 + m_i^2 + \lambda^{1/2}(s, m_j^2, m_i^2)} \right. \\
& \left. + \frac{b}{m_i^2} \frac{m_j^2 - m_i^2 - s}{s^2} \lambda^{1/2}(s, m_j^2, m_i^2) \right],
\end{aligned} \tag{22}$$

and $\lambda(s, m_j^2, m_i^2) = [s - (m_i + m_j)^2][s - (m_i - m_j)^2]$. In the argument list of $\mathcal{R}$, s, m_i, m_j specify the kinematics and quark masses, while a and b are the mass-polynomial coefficients produced by the Dirac trace and the chosen interpolating current; their explicit values are

shown in Eqs. (21) and (24). They have dimensions $[a] = \text{GeV}$ and $[b] = \text{GeV}^3$, so both terms in $\mathcal{R}$ have dimension GeV . Consequently $ds \rho_{A,B}(s)$ has dimension GeV^3 , matching the condensate and photon-DA terms in $\hat{\Pi}_{A,B}$.

The tensor-current invariant amplitude is written in the same notation as

$$\hat{\Pi}_B = \int_{(m_Q+m_q)^2}^{s_0} ds e^{-s/M^2} \rho_B(s) + \frac{m_Q}{m_Q + m_q} \hat{\Pi}_A^{(\chi\phi_\gamma, \mathcal{A}, \mathcal{B})} + \hat{\Pi}_B^{(3p)}. \quad (23)$$

The second term denotes the tensor two-particle photon-DA family inherited from the axial expression in Eq. (17), namely the $\chi_q\phi_\gamma$ and $\mathcal{A}, \mathcal{B}$ pieces. The perturbative tensor spectral density is

$$\begin{aligned} \rho_B(s) = & e_q \mathcal{R} \left(s, m_q, m_Q, \frac{m_Q m_q}{m_Q + m_q}, \frac{m_q^2 s}{m_Q + m_q} \right) \\ & - e_Q \mathcal{R} \left(s, m_Q, m_q, \frac{2m_Q^2 - m_Q m_q}{m_Q + m_q}, -\frac{m_Q^2 s}{m_Q + m_q} \right). \end{aligned} \quad (24)$$

The last term in Eq. (23), $\hat{\Pi}_B^{(3p)}$, is the matched tensor-current three-particle contribution generated by the background-gluon part of the propagator. In the implementation it is expressed through the same integration domains as Eq. (19), with the tensor kernels obtained from the $\sigma_{\alpha\beta}$ and $x_\alpha\gamma_\beta$ parts of Eq. (16). The compact representation used numerically is

$$\hat{\Pi}_B^{(3p)} = e_q \langle \bar{q}q \rangle e^{-m_Q^2/M^2} \mathcal{I}_{3p}^B(u_0), \quad \mathcal{I}_{3p}^B(1/2) = 5.98 \times 10^{-2}, \quad (25)$$

for the central photon-DA parameters. The corresponding axial integral is $\mathcal{I}_{F_1}(1/2) = 7.03 \times 10^{-2}$.

The formulas above are the analog of the explicit LCSR expressions written by Colangelo, De Fazio, and Ozpineci for the single axial current [19]. The difference in the present work is that the OPE is carried out for both basis currents, J_μ^A and J_μ^B , before the physical rotation in Eq. (5).

After the double Borel transformation in p^2 and P^2 and continuum subtraction, we denote by $\hat{\Pi}_A(M_1^2, M_2^2, s_0)$ and $\hat{\Pi}_B(M_1^2, M_2^2, s_0)$ the invariant QCD amplitudes multiplying the structure in Eq. (12) for the two basis currents. Combining these amplitudes according to Eq. (5), the physical charm-strange form factors g_1 and g_2 for $D_{s1}(2460)$ and $D_{s1}(2536)$ are

$$\begin{aligned} g_1 &= \frac{m_c + m_s}{m_{D_s}^2 f_{D_s} m_1 f_1} \exp \left(\frac{m_1^2}{M_1^2} + \frac{m_{D_s}^2}{M_2^2} \right) \left[\sin \theta \hat{\Pi}_A + \cos \theta \hat{\Pi}_B \right], \\ g_2 &= \frac{m_c + m_s}{m_{D_s}^2 f_{D_s} m_2 f_2} \exp \left(\frac{m_2^2}{M_1^2} + \frac{m_{D_s}^2}{M_2^2} \right) \left[\cos \theta \hat{\Pi}_A - \sin \theta \hat{\Pi}_B \right], \end{aligned} \quad (26)$$

TABLE I. Main input parameters. Photon-DA normalizations are quoted at $\mu = 1$ GeV.

Quantity	Value	Comment
m_c, m_b, m_s	1.27(2), 4.18(3), 0.093(11) GeV	$\overline{\text{MS}}$ inputs
$M_{2460}, M_{2536}, m_{D_s}$	2.4595, 2.53511, 1.96835 GeV	PDG
M_{5830}, m_{B_s}	5.82870, 5.36692 GeV	PDG/CMS
f_{D_s}, f_{B_s}	0.2499, 0.2303 GeV	FLAG/lattice context
f_1, f_2	0.345(17), $\simeq 0.38$ GeV	charm axial-meson decay constants
$f_{B_{s1},A}, f_{B_{s1}}^T$	0.305(27), 0.285(27) GeV	bottom basis-current inputs
$f_1^{(B_s)}, f_2^{(B_s)}$	0.443[0.384, 0.492], 0.203[0.121, 0.291] GeV	mixed-current diagonalization
$\chi_s(1 \text{ GeV})$	+3.15(30) GeV^{-2}	legacy photon input
$f_{\gamma,s}^\perp(1 \text{ GeV})$	-55.1(1.9) MeV	preferred photon input

where $m_1 = m_{D_{s1}(2460)}$ and $m_2 = m_{D_{s1}(2536)}$. Here f_1 and f_2 are the leptonic decay constants of the corresponding $D_{s1}(2460)$ and $D_{s1}(2536)$ mesons. We use a Stage-3 decay-constant normalization in which $f_1 = 0.345 \pm 0.017$ GeV is used as the working anchor and the same two-point overlap model gives $f_2 \simeq 0.38$ GeV. As a check of the basis normalization, the pure axial-current limit is compared with the QCD sum-rule result of Ref. [21]. For the bottom-strange channel we use the Pullin–Zwicky basis-current inputs $f_{B_{s1},A} = 0.305 \pm 0.027$ GeV and $f_{B_{s1}}^T = 0.285 \pm 0.027$ GeV [22], and determine the physical-current decay constants from the same mixed-current diagonalization model used for the charm-strange normalization.

IV. NUMERICAL ANALYSIS

A. Input parameters and working windows

The main numerical inputs are collected in Table I. Masses and quark masses are taken from the Particle Data Group [23]. The pseudoscalar decay constants are taken from the lattice averages summarized by FLAG [24]. The photon DAs are those of Ball, Braun and Kivel [18]. We compare the older magnetic-susceptibility input [20] with the recent lattice normalization of the leading-twist photon DA [25]; the lattice-normalized scenario is used as our preferred result.

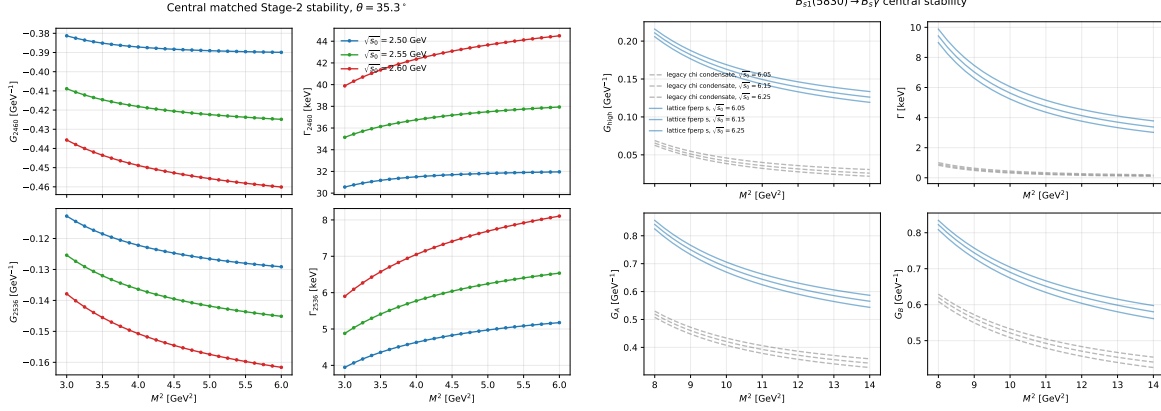


FIG. 1. Representative Borel and continuum-threshold stability checks for the charm-strange and bottom-strange channels.

TABLE II. Final form-factor and width predictions. The lattice photon input is our preferred normalization.

Sector	Decay	Photon input	Normalization	g_X (GeV^{-1})	Γ (keV)
D_s	$D_{s1}(2460) \rightarrow D_s \gamma$	legacy χ_s	$f_1 = 0.344[0.329, 0.363]$	$-0.424[-0.484, -0.360]$	$37.7^{+11.5}_{-10.5}$
D_s	$D_{s1}(2536) \rightarrow D_s \gamma$	legacy χ_s	$f_2 = 0.380[0.339, 0.423]$	$+0.0118[+0.0014, +0.0233]$	$0.045^{+0.124}_{-0.040}$
D_s	$D_{s1}(2460) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$	$f_1 = 0.345[0.330, 0.364]$	$-0.277[-0.312, -0.243]$	$16.1^{+4.4}_{-3.7}$
D_s	$D_{s1}(2536) \rightarrow D_s \gamma$	lattice $f_{\gamma,s}^\perp$	$f_2 = 0.379[0.338, 0.423]$	$+0.0403[+0.0194, +0.0596]$	$0.504^{+0.600}_{-0.377}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	legacy χ_s	$f_2^{(B_s)} = 0.218[0.138, 0.292]$	$-0.0793[-0.130, -0.0537]$	$1.33^{+2.27}_{-0.72}$
B_s	$B_{s1}(5830) \rightarrow B_s \gamma$	lattice $f_{\gamma,s}^\perp$	$f_2^{(B_s)} = 0.203[0.121, 0.291]$	$-0.0401[-0.0695, -0.0274]$	$0.342^{+0.684}_{-0.182}$

The Borel windows are chosen by requiring a stable prediction under changes of M^2 and the continuum threshold while avoiding regions where continuum contributions or higher-dimensional terms dominate. For the charm-strange case we use $3 \text{ GeV}^2 \leq M^2 \leq 6 \text{ GeV}^2$ and $2.50 \text{ GeV} \leq \sqrt{s_0} \leq 2.60 \text{ GeV}$. For the bottom-strange case we use $8 \text{ GeV}^2 \leq M^2 \leq 14 \text{ GeV}^2$. The representative stability plots are shown in Fig. 1.

B. Results and uncertainties

The final form factors and widths are listed in Table II. We quote medians and 16–84 percentile intervals. This is preferable to a Gaussian fit for cancellation-sensitive channels, because the distributions of $D_{s1}(2536)$ and $B_{s1}(5830)$ are visibly non-Gaussian near zero.

The dominant uncertainty in the legacy scenario comes from the photon susceptibility

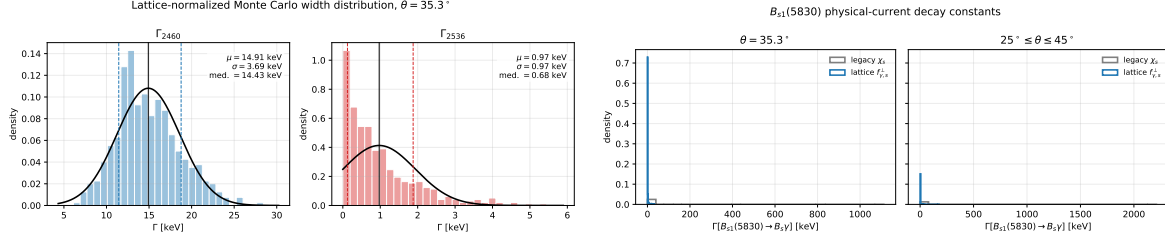


FIG. 2. Monte Carlo width distributions. Left: charm-strange lattice-normalized photon input. Right: bottom-strange physical-current normalization.

normalization and the condensate combination entering $\chi_s \langle \bar{s}s \rangle$. Replacing this input by the lattice-normalized $f_{\gamma,s}^\perp$ reduces the photon-normalization uncertainty and is the reason why we regard the lattice rows as the preferred modern prediction. The mixing angle is the other important source of uncertainty, especially for the cancellation-sensitive $D_{s1}(2536)$ and $B_{s1}(5830)$ amplitudes. The Monte Carlo distributions for the preferred charm and bottom scenarios are shown in Fig. 2.

The suppression pattern can be seen directly at the amplitude level. With the lattice photon normalization, the normalized pieces of the $D_{s1}(2460)$ amplitude are -0.0715 and -0.205 , and therefore add constructively. For $D_{s1}(2536)$ the corresponding pieces are -0.0917 and $+0.132$, so the physical amplitude is a small difference. This is the LCSR counterpart of the cancellation mechanism emphasized in Refs. [14, 15].

C. Comparison with other studies and experiment

Table III compares our preferred values with representative predictions in the literature. The comparison includes early quark-model estimates, relativistic quark-model calculations, recent potential-model or electric-transition analyses, and the phenomenological results of Bondar and Milstein. We compute the pseudoscalar final-state channels $D_s\gamma$ and $B_s\gamma$. The vector-final-state channels $D_s^*\gamma$ and $B_s^*\gamma$ require separate LCSR correlators and are not part of this paper.

For $D_{s1}(2460)$, using the measured branching fraction $\mathcal{B}[D_{s1}(2460) \rightarrow D_s\gamma] = (16 \pm 4 \pm 3)\%$ [1] with our preferred partial width implies a total-width scale of order 0.10 MeV, compatible with a very narrow state. For $D_{s1}(2536)$, the measured total width $\Gamma_{\text{tot}} = 0.92 \pm 0.05$ MeV [2] implies a radiative branching fraction of order 5×10^{-4} in our preferred

TABLE III. Comparison with representative theoretical predictions. Widths are in keV.

Channel	Quark models	Recent QM	Bondar–Milstein	This work
$D_{s1}(2536) \rightarrow D_s \gamma$	15, 25.2–31.1	18.18–18.85	12	0.504[0.127, 1.104]
$D_{s1}(2460) \rightarrow D_s \gamma$	6.2, 10.3–17.2	3.53–3.61	297	16.1[12.4, 20.5]
$B_{s1}^{\text{low}}(5750) \rightarrow B_s \gamma$	37, 70.6	97.7	47	18.4[12.5, 28.1]
$B_{s1}(5830) \rightarrow B_s \gamma$	27, 47.8	56.6	28	0.342[0.160, 1.026]

scenario. This is far below the percent level and naturally explains why the decay is difficult to isolate experimentally. For $B_{s1}(5830)$ there is not yet a direct radiative measurement, and our result should be read as a prediction for future searches.

V. CONCLUSION

We have calculated the radiative transitions $D_{s1}(2460) \rightarrow D_s \gamma$, $D_{s1}(2536) \rightarrow D_s \gamma$, and $B_{s1}(5830) \rightarrow B_s \gamma$ using light-cone QCD sum rules. The calculation includes hard photon emission, two- and three-particle photon DAs, and explicit strange-quark-mass terms. We separated the basis-current sum rules from the physical-meson decay-constant normalization and used a lattice-normalized photon DA as the preferred modern input.

The main result is the strong hierarchy between the two charm-strange radiative widths. The $D_{s1}(2460)$ transition is naturally at the tens-of-keV level, while $D_{s1}(2536) \rightarrow D_s \gamma$ is strongly suppressed by destructive interference between the two physical-current components. The same cancellation-sensitive pattern appears for the observed $B_{s1}(5830)$. In the preferred lattice-normalized scenario we predict a sub-keV median radiative width.

Appendix A: Photon distribution amplitudes

For the nonperturbative photon matrix elements we use the photon DAs in the notation of Rohrwild [20], which is based on the BBK classification [18]. We quote the functions needed in the sum-rule expressions of Sec. III. The variable u is the quark momentum fraction, $\bar{u} = 1 - u$, and $\alpha_q + \alpha_{\bar{q}} + \alpha_g = 1$ for the three-particle DAs.

The leading-twist two-particle DA is taken in the asymptotic form

$$\phi_\gamma(u) = 6u\bar{u}. \quad (\text{A1})$$

The twist-3 vector DA entering the axial invariant amplitude is

$$\psi^{(v)}(u) = -20u\bar{u}(2u-1) + \frac{15}{16}(\omega_\gamma^A - 3\omega_\gamma^V)u\bar{u}(2u-1)[7(2u-1)^2 - 3]. \quad (\text{A2})$$

For the twist-4 two-particle DAs we use

$$\begin{aligned} \mathcal{A}(u) = & 40u\bar{u}(3\kappa - \kappa^+ + 1) + 8(\zeta_2^+ - 3\zeta_2)\left\{u\bar{u}(2 + 13u\bar{u}) \right. \\ & \left. + u^3(10 - 15u + 6u^2)\ln u + \bar{u}^3(10 - 15\bar{u} + 6\bar{u}^2)\ln \bar{u}\right\}, \end{aligned} \quad (\text{A3})$$

$$\mathcal{B}(u) = 40(1 + 3\kappa^+)\int_0^u d\alpha(u-\alpha)\left[-\frac{1}{2} + \frac{3}{2}(2\alpha-1)^2\right]. \quad (\text{A4})$$

The twist-4 three-particle DAs entering Eq. (20) are

$$\begin{aligned} \mathcal{S}(\alpha_i) = & 30\alpha_g^2[(\kappa + \kappa^+)(1 - \alpha_g) + (\zeta_1 + \zeta_1^+)(1 - \alpha_g)(1 - 2\alpha_g) \\ & + \zeta_2\{3(\alpha_{\bar{q}} - \alpha_q)^2 - \alpha_g(1 - \alpha_g)\}], \end{aligned} \quad (\text{A5})$$

$$\begin{aligned} \tilde{\mathcal{S}}(\alpha_i) = & -30\alpha_g^2[(\kappa - \kappa^+)(1 - \alpha_g) + (\zeta_1 - \zeta_1^+)(1 - \alpha_g)(1 - 2\alpha_g) \\ & + \zeta_2\{3(\alpha_{\bar{q}} - \alpha_q)^2 - \alpha_g(1 - \alpha_g)\}], \end{aligned} \quad (\text{A6})$$

$$\mathcal{T}_1(\alpha_i) = -120(3\zeta_2 + \zeta_2^+)(\alpha_q - \alpha_{\bar{q}})\alpha_{\bar{q}}\alpha_q\alpha_g, \quad (\text{A7})$$

$$\mathcal{T}_2(\alpha_i) = 30\alpha_g^2(\alpha_q - \alpha_{\bar{q}})\left[\kappa - \kappa^+ + (\zeta_1 - \zeta_1^+)(1 - 2\alpha_g) + \zeta_2(3 - 4\alpha_g)\right], \quad (\text{A8})$$

$$\mathcal{T}_3(\alpha_i) = -120(3\zeta_2 - \zeta_2^+)(\alpha_q - \alpha_{\bar{q}})\alpha_{\bar{q}}\alpha_q\alpha_g, \quad (\text{A9})$$

$$\mathcal{T}_4(\alpha_i) = 30\alpha_g^2(\alpha_q - \alpha_{\bar{q}})\left[\kappa + \kappa^+ + (\zeta_1 + \zeta_1^+)(1 - 2\alpha_g) + \zeta_2(3 - 4\alpha_g)\right]. \quad (\text{A10})$$

In the preferred modern scenario we replace the legacy normalization $\chi_s\langle\bar{s}s\rangle$ of the leading-twist matrix element by the lattice-normalized $f_{\gamma,s}^\perp$ input. The shapes in this appendix are otherwise kept fixed in the same Rohrwild/BBK convention. When comparing directly with the single-current formula of Ref. [19], these definitions are translated into the equivalent Colangelo notation used in that benchmark expression; no additional DA model is introduced.

TABLE IV. Photon-DA parameters used at $\mu = 1$ GeV in Rohrwild notation.

Parameter	Value	Comment
χ_s	$3.15 \pm 0.30 \text{ GeV}^{-2}$	legacy susceptibility input
$f_{3\gamma}$	$-(4.0 \pm 2.0) \times 10^{-3} \text{ GeV}^2$	twist-3 coupling
ω_γ^A	-2.1 ± 1.0	twist-3 shape
ω_γ^V	3.8 ± 1.8	twist-3 shape
κ	0.15	twist-4 shape
κ^+	-0.05	twist-4 shape
ζ_1	0.40	twist-4 shape
ζ_1^+	0	twist-4 shape
ζ_2	0.30	twist-4 shape
ζ_2^+	0	twist-4 shape

ACKNOWLEDGMENTS

To be added.

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